AP Precalculus — Unit 3, Lesson 3.6

Exit Ticket — Answer Key

Name: ____________________________ Date: ________________ Period: _________

Exit Ticket — Key

1. Identify the amplitude, period, phase shift, and midline of the function

$$f(\theta) = -3\sin\left(2\left(\theta + \frac{\pi}{4}\right)\right) - 1$$

Amplitude: $|-3| = 3$

Period: $2\pi/2 = \pi$

Phase shift: $-\pi/4$ **(shifted left $\pi/4$ units)**

Midline: $y = -1$

2. **(B)** $y = 4\sin(2\theta) + 2$
Period check: $b = 2$ gives period $= 2\pi/2 = \pi$. Amplitude $= 4$. Midline $y = 2$. No phase shift.

3. **The amplitude measures how far the function's outputs rise above (or fall below) the midline (a vertical measurement), while the period measures the horizontal length of one complete cycle.**